

Insulin Dose–Response Alignment for Glucose Forecasting in Type 1 Diabetes

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ABSTRACT

Supervised models trained on observational clinical logs often learn spurious correlations. Held-out prediction error can remain low while the fitted mapping contradicts expected pharmacology, such as predicting that insulin raises glucose. A *dose–response curve* (DRC) defines expected physiological behaviour when a treatment dose is varied while holding other factors fixed. In a fitted model, this response is an *individual conditional expectation* (ICE). Existing explainable artificial intelligence methods can diagnose an unphysiological ICE but cannot repair it.

We introduce DOCAS (dose–response curve alignment via synthetic augmentation). Given a target DRC, DOCAS adds synthetic *do*(·) instances labelled with the curve’s incremental effect and trains the chosen architecture once on the combined data. Alignment error (AE) is the root-mean-square gap between each test-context ICE and the target DRC.

In type 1 diabetes glucose forecasting, the established UVa/Padova simulator supplies the insulin DRC that we use as the target. Across OhioT1DM ($n=12$) and AZT1D ($n=24$), DOCAS brings the test-set insulin ICEs into line with that DRC at mean RMSE increases of 2.7–4.4%. In an in-silico decision-support evaluation, an aligned model increases post-decision time in range from 73.61% to 76.01% without inducing hypoglycaemia under an explicit safety guard. These results indicate that aligning the treatment response is as important as minimizing prediction error when a forecaster is used for dosing decisions.

1. Introduction

Glucose forecasters trained on observational type 1 diabetes data often predict that more insulin on board leads to higher future glucose [1–4]. This is caused by confounding in how insulin is administered: patients usually bolus when their glucose is already high or when they eat. In the logs, large insulin doses therefore appear together with high and rising glucose, and a model that minimizes the prediction error learns this association instead of the effect of the dose [5, 6]. The held-out error does not reveal this problem, since the test data follow the same pattern. The problem only becomes apparent when the model is asked about a dose that was not given in that situation, for example a larger correction bolus. In this case the model extrapolates the learned association and predicts that glucose will rise.

In clinical practice, such a model can lead to both underdosing and overdosing. Too little insulin leaves glucose high, and prolonged insulin deficiency increases the risk of diabetic ketoacidosis as well as the long-term risk of microvascular complications. Too much insulin, on the other hand, lowers glucose too far. Hypoglycaemia can impair cognitive function and, in severe episodes, lead to seizures or loss of consciousness. A forecaster that has learned the wrong sign tends to reduce or withhold a correction bolus and thus underdoses, whereas a forecaster that overestimates the glucose drop recommends a larger bolus than needed and thus overdoses.

For a dosing decision, the relevant question is how glucose would change if a different dose were given in the same situation. Answering this question requires a *ceteris paribus* comparison. The Latin phrase means “all other things being equal”: only one factor is changed while everything else is kept as it was, so that any difference in the outcome can be attributed to this factor alone. For insulin, this means comparing the glucose that would follow different doses at the same current glucose, the same meal, and the same time of day. Pearl’s *do*(·) operator provides a notation for this comparison [5, 7]. Observing that a patient took u units is not the same as setting the dose to u , since the observation

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also contains the reason why the dose was chosen, e.g. high glucose. In contrast, $do(\text{insulin}=u)$ means that the dose is actively set to u while nothing else is changed. The logs only contain observations and therefore mix the effect of insulin with the reason it was given.

Relation between the dose–response curve and ceteris paribus plots in machine learning. The *dose–response curve* (DRC) describes the expected change in glucose under $do(\text{insulin}=u)$ as a function of the dose u . For insulin, this curve is well characterized at the population level: the more insulin, the larger the drop in glucose. Metabolic simulators such as UVa/Padova describe this curve and also quantify how insulin sensitivity varies between people [8, 9]. The same question can also be asked of a fitted glucose forecaster, and the resulting curves are known in machine learning as *ceteris paribus* plots [10]. To obtain such a curve, one takes a single situation from the data, i.e. one row of model inputs such as current glucose, recent carbohydrates, and insulin on board. All inputs except insulin on board are kept fixed, insulin on board is varied from zero to a high dose, and the predicted glucose change is recorded at each value. The resulting curve is called *individual conditional expectation* (ICE) curve [11]. Here, “individual” refers to this one situation and not to a patient-specific pharmacology, so a test set with many situations yields many ICE curves. Averaging them gives the *partial dependence plot* (PDP) [12]. Both plots have their advantages. The PDP is a single curve and therefore easier to read when the question is how the forecaster responds to insulin on average. An ICE curve, in contrast, belongs to one context row, so a set of ICE curves shows how the response varies between situations, including situations in which the model responds in opposite directions that are hidden in the average. Note that the DRC and these plots do not share the same y-axis. The DRC is the observed drop in glucose when a dose is taken, whereas ICE curves and PDPs show the change in the model’s prediction when only the dose input is changed. A decision support system that chooses a bolus for one patient at one point in time relies on the ICE curve of that situation. *Held-out accuracy does not test the insulin response.* Glucose forecasters are usually validated by their prediction performance on held-out data, most often measured by the root-mean-square error (RMSE). However, a model can perform well in terms of RMSE and still respond to insulin with the wrong sign. ICE curves and PDPs make this response visible, and many studies use them for this purpose. In most cases, however, the interpretation remains visual and qualitative, since there is no reference curve against which the plots could be compared. Type 1 diabetes is different in this regard, as the population insulin dose–response is already well characterized and can serve as such a reference. Each test-context ICE can then be compared with this curve via the root-mean-square distance and reported next to the RMSE. In this way, a forecaster that predicts glucose well but responds wrongly to insulin can be identified quantitatively and not only by visual inspection.

Motivation for the study. Insulin pharmacodynamics in type 1 diabetes are well characterized, and public datasets with glucose, meal, and insulin logs are available. Prendin et al. showed that a neural glucose forecaster with competitive accuracy responded to insulin in the wrong direction, and that only a forecaster with a plausible insulin response improved glycaemic control when used as a corrective-bolus decision support system (DSS) [1]. Such systems rely on data-driven forecasters that are trained on the patient’s logs and selected based on their prediction performance. Therefore, we aimed to correct the insulin response of the forecaster rather than to replace it or to change its architecture. Architectural constraints tie the correction to one model family, and monotonic constraints only fix the sign of the response (Section 2). Instead, we add the insulin dose–response to the training data. Each synthetic row is a copy of a real situation in which insulin is set to a new dose, as in $do(\text{insulin}=u)$, and the glucose change is labelled according to the target DRC. The learner is then trained once on the union of observed and synthetic rows. In this way, the population curve determines the insulin response while the patient’s own observations shape the rest of the model.

Objectives of this paper. (i) We define alignment error as an evaluation metric for the insulin response of a glucose forecaster, reported alongside RMSE. (ii) We introduce DOCAS (dose–response curve alignment via synthetic augmentation), which aligns that response through synthetic training rows while the forecaster itself stays unchanged. (iii) We evaluate DOCAS on OhioT1DM and AZT1D with CatBoost and a neural network. (iv) We test the aligned forecasters in silico as a corrective-bolus decision support system. (v) We study how many synthetic rows are needed and compare training on them with applying the DRC after prediction.

2. Related Work

Post-hoc model explanations. SHAP and LIME decompose individual predictions into additive feature contributions [13, 14]. They identify which inputs influenced a factual prediction, but they do not characterize how the prediction changes when a treatment dose is set while other factors remain fixed. Saliency sanity checks and fidelity metrics ask whether an explanation matches the predictor’s internal logic [15–18]. Causal SHAP formulations quantify interventional feature

importance without enforcing a continuous functional shape [7, 19, 20]. All of these tools inspect a fitted model, so they can reveal a wrong-signed insulin effect [1, 4, 21] but leave the predictor unchanged.

Ceteris paribus: ICE and PDP. Individual conditional expectation (ICE) and partial dependence plots (PDP) (Section 1) are the standard *ceteris paribus* views of a fitted model [11, 12]. An ICE traces the prediction as one input is swept and the remaining context is held fixed, so there is one ICE per instance; a PDP averages those curves. In clinical decision support the relevant object is the ICE of the context at which a dose is chosen, because averaging in a PDP can hide opposing slopes. Because training logs can be biased in the way described above, the ICEs on a test set should be scored against pharmacological knowledge, for example an established DRC, and not only by held-out error.

Explanation-guided training. Ross et al., Erion et al., and Rieger et al. penalize input gradients, expected gradients, or spatial saliency so that attributions follow a stated importance pattern [22–25]. Because they align the explanation rather than the dose–response curve, a correct attribution sign can coexist with a flat or inverted ICE, so regularizing feature importance does not guarantee a physiological what-if response.

Synthetic data augmentation. A second family adds extra training rows rather than changing the learner. Zhang et al. interpolate between real examples and their labels [26]. Kaushik et al. copy an instance and change a spurious attribute while holding the rest fixed, which reduces shortcut use [27]. Both remain model-agnostic, yet their labels still follow the observational process, or an interpolation of it, not a stated interventional DRC.

Simulator-generated trajectories. In T1D, generative models synthesize extra glucose time series, sometimes inside a digital-twin workflow, so that a learner can be pretrained or trained on a larger cohort [28–30]. A digital twin here is the concept of adapting a physiological simulator to an individual, not a model class. The added labels are simulated glucose trajectories rather than a sweep over insulin doses with everything else fixed, so the learner sees more data but is not told what the dose–response should be.

Monotonicity and shape constraints. Split constraints in boosted trees and architectural constraints in neural networks enforce a non-increasing or non-decreasing map from a treatment feature to the prediction [31, 32]. In glucose–insulin models a conformance constraint can further bound the size of the insulin effect [2]. These methods specify the sign of the response, and sometimes a magnitude cap, but not a physiological curvature.

Hybrid first-principles models and physics-informed networks. Hybrid semi-parametric models combine a parametric first-principles block with a nonparametric data-driven block [33]. The equations remain part of the model at prediction time, and this structure also allows them to extrapolate beyond the data [34, 35]. In serial designs a network estimates kinetic parameters that are passed on to balance equations, so the learner itself is not changed. Physics-informed neural networks instead add the residuals of known differential equations to the training loss [36]. In biomedicine the equations usually sit in a dedicated simulator, and a digital twin is the concept of adapting such a simulator to one person rather than a model class of its own [3, 9].

Interchange-intervention training. Esponera and Cinà regularize hidden units, under swapped inputs, against a glucose simulator [37], so they align internal latent structure rather than the input–output treatment ICE used for what-if dosing.

Table 1 summarizes these alignment approaches. Note that the list by no means claims to be exhaustive, and other approaches exist. None of the listed approaches reports the distance between per-context treatment ICEs and a stated dose–response curve as a metric, and none aligns those ICEs while leaving a standard supervised learner unchanged.

Table 1

Approaches that align a supervised model to a stated behaviour, grouped by *what* the model is aligned to. Model agnostic: the same standard supervised learner can be used without a custom layer, loss, or mechanistic core. Example papers are representative, not exhaustive.

Approach	Example papers	Model agnostic?	Aligned to	Domain
Explanation-guided training	Ross [22]; Rieger [24]; Erion [23]	No	Feature attributions, not the dose–response.	Decoy images and text [22]; medical imaging [24]; genomics [23]
Synthetic data augmentation	Zhang [26]; Kaushik [27]	Yes	Extra-row labels, not an interventional dose–response curve.	Image classification [26]; natural language processing [27]
Simulator-generated trajectories	Noguer [28]; Zhu [29]; Prendin [30]	Yes	Simulated trajectories, not a dose sweep.	Type 1 diabetes glucose time series [28–30]
Monotonicity and shape constraints	Ke [31]; Runje [32]; Kushner [2]	No	Sign of the treatment effect, not its curvature.	Tabular learning [31, 32]; type 1 diabetes glucose–insulin [2]
Hybrid first-principles models	von Stosch [33]; Fiedler [34]	No	First-principles block kept in the model.	Process systems engineering [33, 34]
Physics-informed networks	Raissi [36]	No	Partial or ordinary differential equation residuals in the loss.	Forward and inverse partial differential equations [36]
Interchange-intervention training	Esponera [37]	No	Hidden units, not the treatment individual conditional expectation.	Type 1 diabetes glucose forecasting [37]

3. Methods

All code to reproduce the experiments is publicly available on GitHub (<https://github.com/Prgrmmrjns/DOCAS-manuscript>). We run DOCAS in three phases (Figure 1): define the target DRC that specifies the expected response under $do(\cdot)$ (Phase 1), add synthetic counterfactual training rows labelled with that mapping (Phase 2), and train the chosen supervised forecaster on the combined dataset (Phase 3). This paper applies the three phases to insulin in type 1 diabetes.

3.1. Conceptual formulation

A supervised model is trained on observational records, each of which pairs a context—all input features other than the treatment—with the dose that was actually given and a continuous outcome. Empirical risk minimization fits the observational association between those inputs and the outcome. When higher doses are reserved for more severe cases, that association can run opposite to the drug’s biological effect: more treatment then predicts a worse outcome. Figure 1A shows this for insulin: in the observed data, future glucose rises with insulin on board (red dashed line), whereas the target DRC (dotted line) falls.

Decision support asks a different question: it requires the prediction that would follow from setting the treatment while holding the context fixed. Sweeping the treatment feature across a dose range, with all other features of that record unchanged, yields an individual conditional expectation (ICE): one curve per context, reported as the change in prediction relative to the value at null dose [11]. Dose is normalised to the unit interval using a domain-specific maximum. A partial dependence plot (PDP) averages these curves over many contexts and can hide opposing slopes.

The *target DRC* is the expected dose–response from mechanistic principles or clinical consensus. It is a stated prior, not patient-specific ground truth, because the true interventional response is not observed in static logs. *Alignment error* (AE) is the root-mean-square gap between each of those ICEs and the target DRC, taken over all evaluation contexts and a dense grid of doses:

$$AE = \sqrt{\text{mean}[(ICE - DRC)^2]}. \quad (1)$$

AE uses the physical units of the outcome, while held-out RMSE measures factual accuracy on observational test data.

3.2. Generic DOCAS framework

Phase 1: defining the target DRC and context. We define the target DRC as the expected change in outcome when treatment is set to a given dose, holding the remaining inputs—the context—fixed. Because observational datasets

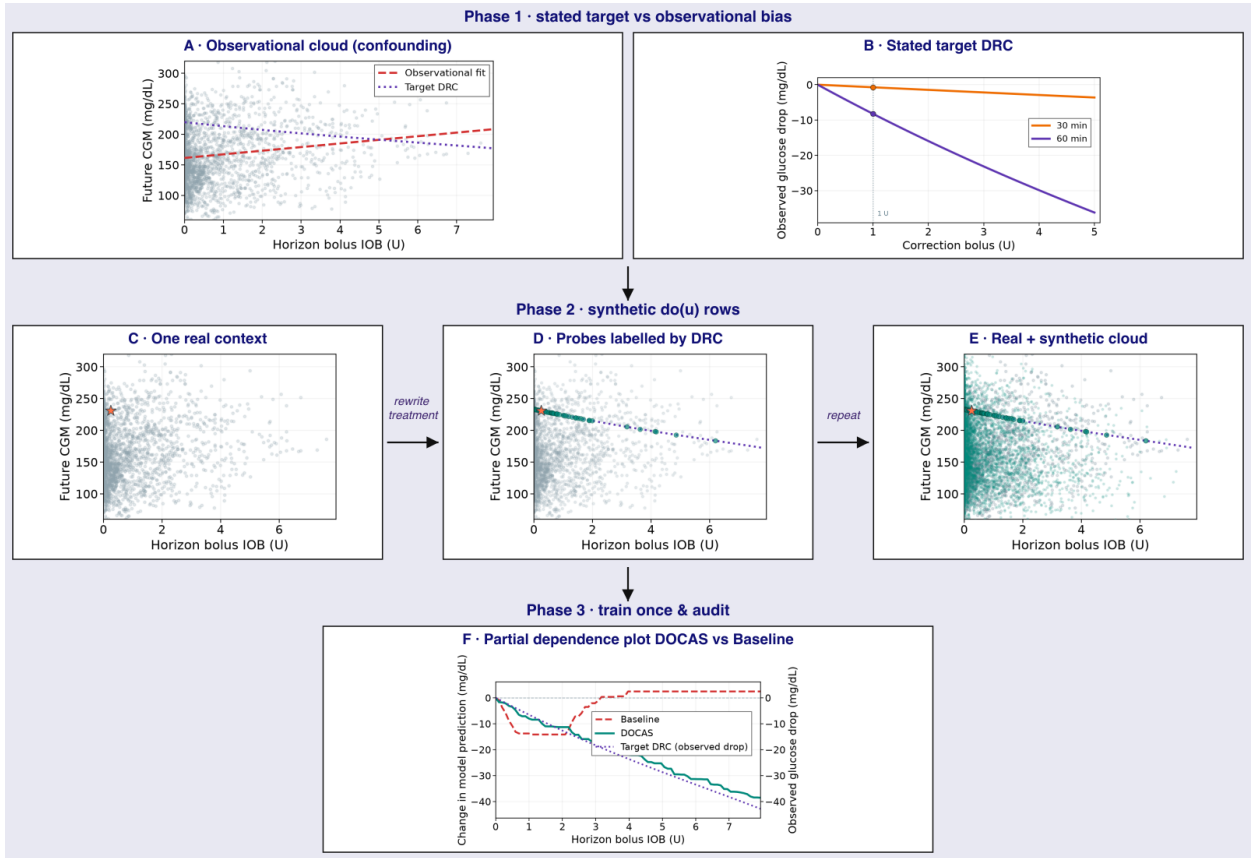


Figure 1: Overview of the DOCAS alignment procedure across three phases (illustrated on OhioT1DM patient 588). **Phase 1.** **A:** observational training distribution exhibiting treatment-indication confounding, contrasted with the slope of the target DRC. **B:** stated target DRC at 30 and 60 minutes. Its vertical axis is the observed glucose drop when that dose is taken. **Phase 2.** **C–E:** counterfactual instance generation: copying observed contexts, sweeping the treatment feature across probe doses, and assigning labels via the factual outcome plus the incremental target DRC effect. **Phase 3.** **F:** partial dependence plot (mean ICE across test contexts) of DOCAS versus Baseline (original training records only). Its vertical axis is the change in the model's prediction. A second vertical axis shows the target DRC, the observed drop those predictions are compared with.

rarely cover wide dose sweeps under uniform contexts, the target DRC supplies the missing interventional supervision. Figure 1B shows the target DRC used in this study, i.e. the observed glucose drop 30 and 60 minutes after a correction bolus of a given size.

Phase 2: generating synthetic counterfactual instances. In Phase 2, DOCAS creates additional training rows that show the model how the outcome should change when only the dose changes. The observed rows are kept as they are. The number of synthetic rows is set as a multiple of the number of observed training rows, which we call the synthetic budget; a budget of 20, for example, means twenty synthetic rows per observed row. Each synthetic row is created in four steps.

First, a real training row is drawn at random, with replacement, and copied. It provides a realistic context, i.e. a combination of current glucose, recent carbohydrates, and the other inputs that actually occurred in the data (star in Figure 1C).

Second, a small amount of random noise is added to the continuous context features. In the type 1 diabetes application, we add Gaussian noise with a standard deviation of 10 to current glucose (in mg/dL) and to recent carbohydrates (in g), keeping glucose above 40 mg/dL and carbohydrates non-negative. Since each observed row is copied many times,

the noise prevents the model from seeing the same context over and over again and spreads the synthetic rows slightly around the observed contexts.

Third, the dose in the copied row is replaced by a new probe dose, which is sampled in one of two ways. Half of the probe doses are drawn from the doses that actually occur in the training data (density-based sampling). Doses that are common in practice, such as small correction boluses, are therefore sampled often, so the model learns the response well where most dosing decisions are made. The other half are drawn uniformly between zero and the maximum dose (uniform sampling). This ensures that large doses, which are rare in the logs, are also covered, so that the model learns the response over the whole dose range and not only where the data are dense. The green points in Figure 1D are such probe doses for the copied context.

Finally, each synthetic row receives a label. We take the outcome that was actually observed for the copied row and add the difference in the target DRC between the probe dose and the dose that was actually given. For example, if the target DRC states that the probe dose lowers glucose by 15 mg/dL more than the dose that was given, the label is the observed glucose change minus 15 mg/dL. In this way, everything apart from the dose that influenced the observed outcome, e.g. a meal or sensor noise, remains in the label, while the effect of the dose change follows the target DRC. Therefore, the labelled probes in Figure 1D lie on the target DRC that passes through the copied context. Repeating these steps for many randomly drawn contexts yields the combined cloud of real and synthetic rows in Figure 1E.

Phase 3: model training and evaluation. The synthetic rows are pooled with the observational data, and the same supervised model is trained once on the combined set, using its native loss, with no extra penalty terms and no change to the architecture. Performance is evaluated on held-out test data via RMSE and via AE on the ICE of every test context (Eq. 1). Figure 1F compares the resulting partial dependence plot of DOCAS with the baseline trained on the observed rows only. The left axis shows the change in model prediction, and the right axis shows the target DRC as observed glucose drop; the baseline curve is flat or rises at higher doses, while the DOCAS curve follows the target DRC.

3.3. Application to type 1 diabetes

We instantiate DOCAS for postprandial glucose forecasting on OhioT1DM (open-loop pump therapy) and AZT1D (closed-loop automated insulin delivery; cohort protocol in Section 3.4), with bolus insulin on board (IOB) as the treatment feature, basal insulin left unmanipulated, and the 30- or 60-minute change in continuous glucose monitor (CGM) readings as the outcome.

The target DRC is the interstitial glucose drop of a rapid-acting correction bolus, simulated in the UVa/Padova model as implemented in ReplayBG [3, 38] and scaled by the patient's current glucose so that the same dose lowers glucose more when ambient glycaemia is higher; both cohorts use this construction. Simulation settings, sensor profiles, and the normalised dose axis are given in Appendix A. Figure 2 shows the simulated trajectory and the resulting target DRC.

Synthetic labels keep the observed residual and add the change in the target DRC between the probe dose and the dose that was actually given, at a validation-selected budget of twenty times the training size (Section 4.4).

3.4. Experimental setup and evaluation protocol

Our evaluation adopts the protocol of Prendin et al. [1]: identical OhioT1DM train/test splits, IOB and carbohydrate on board (COB) features, and the ReplayBG in-silico decision-support benchmark. Models map current CGM, bolus IOB, and COB to future CGM change over 30- or 60-minute horizons. For OhioT1DM ($n=12$), the initial six monitoring weeks serve as training data and the final ten days are held out for testing [1, 39]. On-board features use causal event histories available at forecast time.

We evaluate performance across cohorts via mean $\pm$ standard deviation of held-out RMSE and alignment error (AE) across all test contexts and a dense dose grid from none to the domain maximum (Eq. 1). The primary model architecture is CatBoost [40]. Baseline is the same learner trained only on the original training records; DOCAS is trained on those records plus the synthetic rows of Phase 2 (Appendix A, 20 $\times$ budget, 50%/50% empirical/uniform mix). Appendix B repeats the comparison for a multilayer perceptron (MLP) and for CatBoost with 12 five-minute lags per channel (36 inputs).

The AZT1D dataset [41, 42] provides an external validation cohort of 24 participants on automated insulin delivery (Tandem t:slim X2 Control-IQ with Dexcom G6 Pro). Subject 14 is excluded due to lack of an isolated CGM channel. We employ an 80/20 chronological split per patient, matching feature definitions, the target DRC of Appendix A, and evaluation metrics.

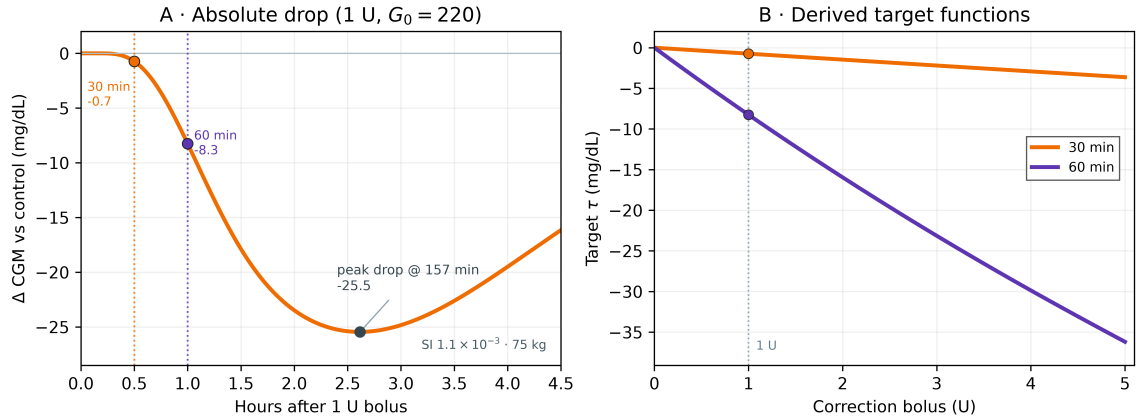


Figure 2: Ambient-scaled target DRC derived from ReplayBG simulation. **A:** interstitial glucose reduction following a 1 U bolus versus no-bolus control at $G_0=220$ mg/dL, indicating the drop at 30- and 60-minute horizons. **B:** derived target functions versus correction bolus across prediction horizons (1 U reference marked at baseline ambient glycaemia).

Statistical significance between baseline and DOCAS arms is evaluated using paired, two-sided Wilcoxon signed-rank tests across patients with Holm–Bonferroni correction ($\alpha=0.05$). CatBoost employs untuned default parameters (depth 6, automatic learning rate, 1000-iteration maximum) with early stopping (patience 50) evaluated on a chronological 20% validation split of each arm’s training distribution. Primary results in Table 2 and the MLP rows of Appendix B report averages over five random seeds (42–46); the lag-feature rows, ablations, and parameter sweeps use seed 42.

4. Results

4.1. Held-out accuracy and alignment

Figure 3 shows the insulin PDP of each patient, i.e. the mean of the ICE curves over the patient’s test contexts. The baseline CatBoost, trained only on the original records, produced curves that were flat or even increased with insulin, which is the pattern expected from treatment-indication confounding. With DOCAS, the curves decreased with insulin and followed the stated DRC for both horizons. Table 2 gives the cohort averages over five seeds.

In both cohorts, DOCAS reduced the alignment error by 89–93% compared with the baseline, while the RMSE increased by only 2.7–4.4%. On OhioT1DM, AE decreased from 12.70 mg/dL (baseline) to 1.44 mg/dL (DOCAS) at 30 minutes and from 38.13 to 2.79 mg/dL at 60 minutes, with an RMSE increase of 3.2% and 4.4%. For AZT1D, AE decreased from 19.62 to 1.70 mg/dL and from 49.30 to 4.13 mg/dL, with an RMSE increase of 2.7% and 3.0%. Alignment improved for every participant in both cohorts, and the reductions were significant for all cohorts and horizons (paired Wilcoxon tests with Holm–Bonferroni correction). The same was observed for an MLP, and adding lag features reduced the DOCAS alignment error even further (Appendix B).

The results on participant level are consistent with these averages. On OhioT1DM, even the participant with the smallest improvement reduced AE by more than 4 mg/dL at 30 minutes and by almost 14 mg/dL at 60 minutes. Since the synthetic rows are only used for training and AE is evaluated on held-out test contexts, this shows that the learned insulin response generalizes to unseen situations. For dosing, the 60-minute horizon is the more relevant one, since after 30 minutes a bolus has lowered glucose only slightly, often by less than the sensor noise.

The RMSE cost was small for all participants. DOCAS even lowered the RMSE for 2 of 12 OhioT1DM participants at 30 minutes and for 3 of 12 at 60 minutes, and for every participant the increase in RMSE was less than a quarter of the AE improvement. Note that the RMSE measures how well the forecaster predicts glucose when nothing is changed, while the ICE determines which dose it recommends.

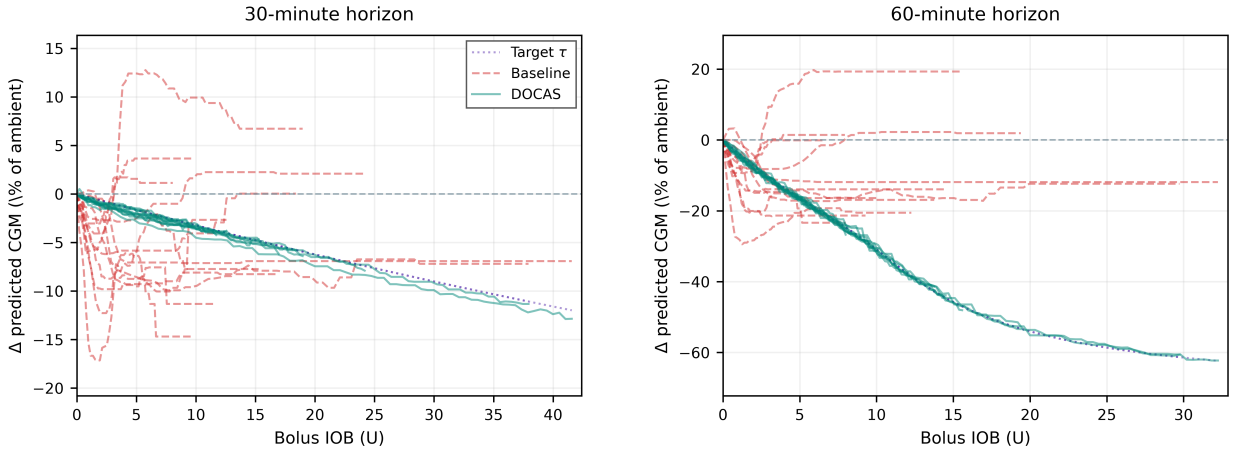


Figure 3: Patient-level insulin PDP on OhioT1DM (mean ICE per patient). Alignment error in Table 2 is evaluated on the per-context ICE versus the target DRC. Δ predicted CGM expressed as % of mean test ambient CGM. **Left:** 30-minute horizon; **right:** 60-minute horizon. Curves extend to each patient's training-maximum IOB.

Table 2

Held-out CatBoost RMSE and alignment error on both cohorts. Baseline: original training records only. DOCAS: original records plus synthetic Phase 2 rows. Cohort mean $\pm$ SD over 5 seeds. Bold*: significantly better arm (Section 3.4). MLP and lag features in Appendix B.

Approach	30 min		60 min	
	RMSE	Align.	RMSE	Align.
OhioT1DM ($n = 12$)				
Baseline	21.91 $\pm$ 3.04*	12.70 $\pm$ 5.14	34.32 $\pm$ 4.35*	38.13 $\pm$ 22.19
DOCAS	22.61 $\pm$ 2.62	1.44 $\pm$ 0.67*	35.83 $\pm$ 3.64	2.79 $\pm$ 0.56*
AZT1D ($n = 24$)				
Baseline	21.60 $\pm$ 3.44*	19.62 $\pm$ 16.11	31.82 $\pm$ 5.55*	49.30 $\pm$ 29.29
DOCAS	22.19 $\pm$ 3.32	1.70 $\pm$ 0.84*	32.78 $\pm$ 5.18	4.13 $\pm$ 2.44*

4.2. Investigating the alignment and prediction performance trade-off

Figure 4 illustrates the trade-off between alignment and prediction performance. Among test hours where DOCAS reduced the 60-minute forecast ($r = -0.50$, $n=14,993$), alignment improved accuracy during glucose drops (mean absolute error improvement 4.49; 88% of instances) and degraded accuracy during postprandial rises (-5.14 ; DOCAS superior in 15%). Participant 544 represents an extreme post-meal state: glucose rises from 148 to 232 mg/dL. The observational model predicts this rise (≈ 213), whereas DOCAS predicts ≈ 158 based on active IOB. Conversely, participant 563 represents an isolated correction without active carbohydrate: glucose falls from 278 to 200 mg/dL, and DOCAS (≈ 207) outperforms the observational baseline (≈ 251).

4.3. In-silico decision support

We integrate each trained model into the corrective-bolus decision support system (DSS) of Prendin et al. [1] and evaluate outcomes in silico using ReplayBG [3]: a physiological simulator fitted to real glucose–insulin trajectories, replaying identical episodes under alternative dosing policies. The protocol evaluates all 12 OhioT1DM participants under matched conditions. AZT1D is omitted from replay because its automated Control-IQ algorithm already issues closed-loop corrections (median 35.6% automatic boluses, 41.4% automated basal modulation), confounding independent DSS evaluation.

The benchmark isolates eight-hour postprandial windows without subsequent carbohydrate or manual boluses. Two hours post-meal, if glucose exceeds 180 mg/dL, the DSS evaluates candidate boluses from 0 to 5 U in 0.5 U increments, minimizing the squared deviation from a 120 mg/dL target at 60 minutes with a quadratic dose penalty [1]. If no dose improves on zero, no bolus is given; a second trigger is permitted after 120 minutes. To address the fact

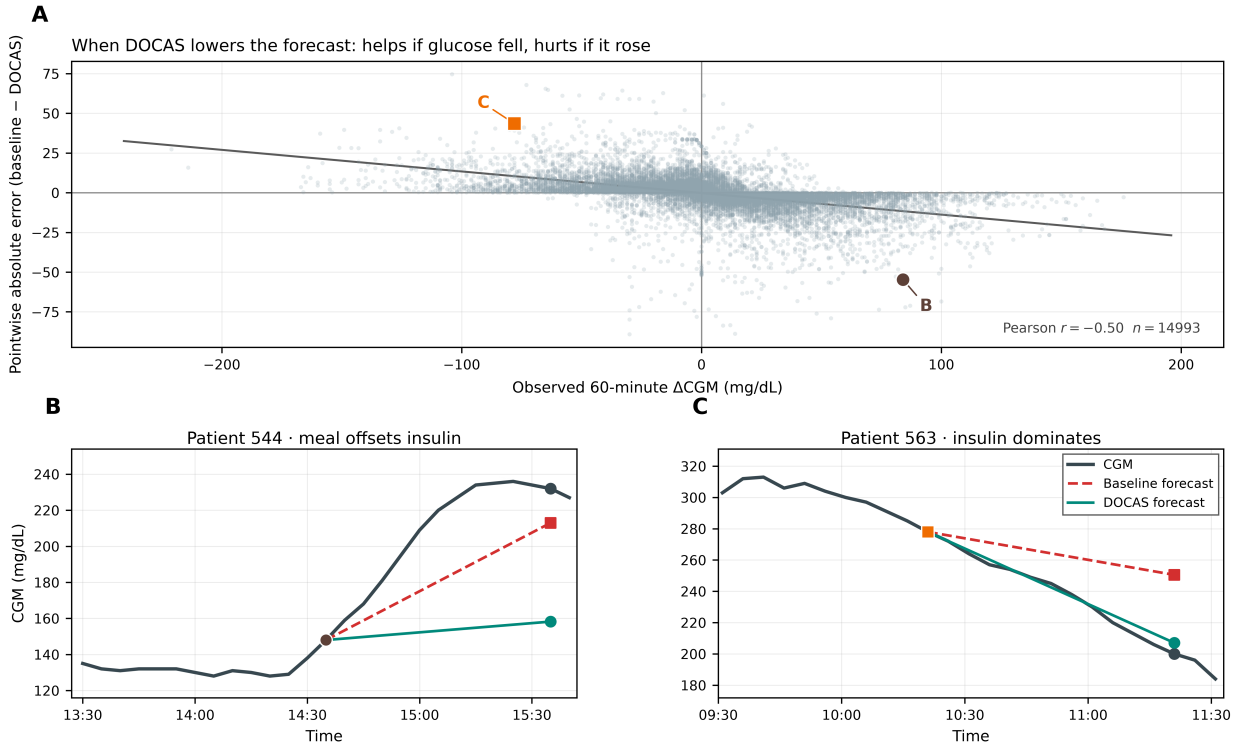


Figure 4: Evaluation of 60-minute predictions under aligned versus observational models. **A:** test hours where DOCAS lowered the prediction: observed Δ CGM versus pointwise absolute-error improvement (baseline minus DOCAS; Pearson $r = -0.50$, $n = 14,993$; markers B and C denote instances in panels B and C). **B:** participant 544 (meal absorption counteracting insulin action). **C:** participant 563 (insulin-dominated glucose decline).

that only $\approx 20\%$ of rapid-acting insulin acts within 60 minutes, we incorporate an explicit hypoglycaemia safety guard (Appendix C, Figure 6): candidates are admissible only if their projected 180-minute nadir remains above 70 mg/dL. Rejected candidates step down to the highest admissible lower dose.

Windows qualify if the twin fits recorded glucose ($\text{RMSE} \leq 25$ mg/dL), baseline replay exhibits hyperglycaemia, and a 1 U probe bolus yields a clinically plausible drop (≤ 100 mg/dL, nadir ≥ 70 mg/dL). Eleven eligible windows across four participants met all criteria; nine triggered active dosing (Table 3).

Post-decision time in range (TIR, 70–180 mg/dL) is 61.49% with no decision support, 73.61% under Baseline (mean dose 1.09 U), and 76.01% under DOCAS (mean dose 2.23 U), with zero time below range (TBR, < 70 mg/dL) across all arms under the safety guard. The paired TIR improvement is +2.40 percentage points (pp) across windows ($p = 0.062$, positive in all five windows where TIR differed). Sweeping the dose penalty confirms that DOCAS maintains higher TIR across operating points: the observational DSS plateaus at 1.41 U due to its upward-sloping ICE, whereas DOCAS scales from 0.73 to 2.32 U. At matched dosing (~ 1.09 U) the arms are comparable, confirming that alignment enables safe, larger therapeutic corrections.

Across the nine triggering decision points, the observational ICE deviates by a mean of 17.27 mg/dL from the target curve, whereas DOCAS deviates by 1.83 mg/dL. At a 5 U probe, the observational model predicts a 7.02 mg/dL glucose *increase*, whereas the target DRC specifies a 25.40 mg/dL drop and DOCAS predicts a 26.39 mg/dL drop. The DOCAS ICE is strictly non-increasing across all nine decisions, compared to once for the baseline. Figure 5 shows one window for OhioT1DM patient 588: the observational ICE turns upward beyond 2.5 U and the DSS picks 0.5 U, whereas DOCAS follows the target DRC and picks 2.0 U.

Table 3

Post-decision outcomes for no decision support and the two corrective-bolus policies on 11 OhioT1DM windows from 4 participants (mean $\pm$ SD).

Metric	No DSS	Baseline	DOCAS
TBR (%)	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
TIR (%)	61.49 $\pm$ 34.22	73.61 $\pm$ 23.83	76.01 $\pm$ 21.95
TAR (%)	38.51 $\pm$ 34.22	26.39 $\pm$ 23.83	23.99 $\pm$ 21.95
Insulin (U)	0.00 $\pm$ 0.00	1.09 $\pm$ 1.41	2.23 $\pm$ 2.04

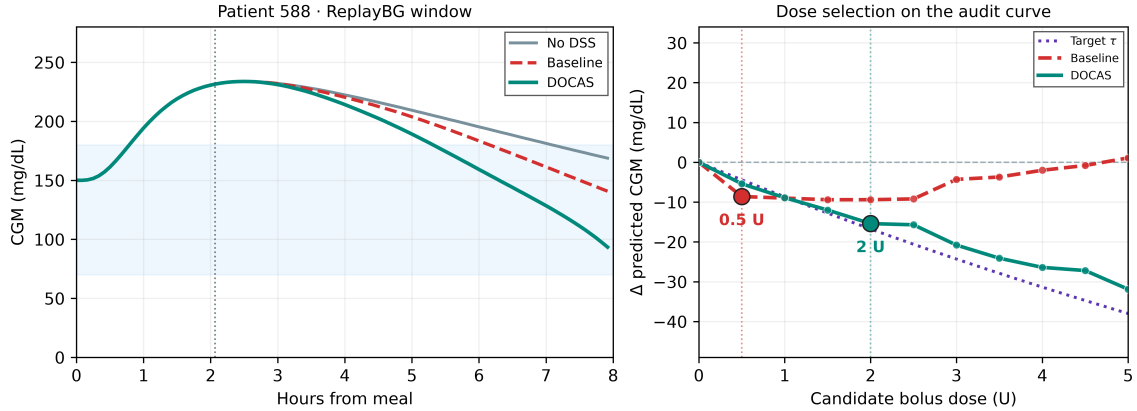


Figure 5: In-silico postprandial replay for OhioT1DM patient 588 (CatBoost, 60-minute horizon). **Left:** simulated glucose trajectories without DSS and under observational versus DOCAS-guided dosing. **Right:** decision-context ICE (marker denotes selected dose). The observational ICE turns upward beyond 2.5 U (0.5 U chosen); DOCAS conforms to the target DRC (2.0 U chosen). Post-decision time in range increases from 13.89% (No DSS) and 30.56% (Baseline) to 44.44% (DOCAS), with zero time below range.

4.4. Ablation studies

We ablate four parts of the method on both cohorts (CatBoost, seed 42; Table 4): how many synthetic rows are added, how probe doses are drawn, whether the target DRC can be added after training instead of during it, and whether the stated curve itself can be changed.

The study preset uses twenty synthetic rows per observed training row (20 $\times$). A 0.1 $\times$ budget already cuts 30-minute AE (9.23 vs. 12.70 mg/dL on OhioT1DM; 8.21 vs. 19.62 mg/dL on AZT1D) at essentially no RMSE cost. At 1 $\times$, AE is 4.62/9.50 and 4.48/8.12 mg/dL. The 20 $\times$ preset reaches 1.51/2.75 and 1.64/3.64 mg/dL; 100 $\times$ tightens AE further (0.82/2.13 and 0.84/2.32 mg/dL) with only a small extra RMSE rise. More synthetic rows therefore pull the per-context ICEs closer to the target, at a small factual cost.

Probe doses in the study mix are half from the training dose density and half uniform on [0, 1]. Drawing every synthetic IOB uniformly instead leaves AE much higher (7.37/11.13 mg/dL on OhioT1DM; 3.31/7.27 mg/dL on AZT1D) at almost the same RMSE, so coverage of common clinical doses matters.

Adding the target DRC after a no-insulin fit, rather than training on synthetic rows, forces AE=0 but raises 60-minute RMSE (39.03 vs. 35.80 on OhioT1DM; 36.81 vs. 32.82 on AZT1D) and cannot change the recommended dose with context, so alignment has to enter during training.

Scaling the prior insulin sensitivity S_I by 0.5 $\times$ or 2 $\times$ keeps AE in the same few mg/dL, so the method follows the stated curve rather than a fixed amplitude. Replacing UVa/Padova DRC with a pump-settings curve (Eq. 3, Appendix D) still aligns, with RMSE 0.31 (OhioT1DM, 30 min) and 0.72 (AZT1D, 60 min) above the 20 $\times$ UVa/Padova run.

5. Discussion

Summary of results. This study aimed to evaluate and correct the insulin response of glucose forecasters trained on observational type 1 diabetes data. On OhioT1DM and AZT1D, DOCAS brought the insulin ceteris paribus plots close to the target DRC at a small RMSE cost, for every participant and for both CatBoost and an MLP (Table 2, Appendix B).

Table 4

Ablations on both cohorts (mean $\pm$ SD, seed 42). Synthetic budgets 0.1 $\times$, 1 $\times$, and 100 $\times$ versus the study preset 20 $\times$. Uniform IOB only draws every probe from Uniform[0, 1] (no training-density mix). The last row uses a pump-settings DRC (Section 4.4).

Ablation	OhioT1DM ($n = 12$)				AZT1D ($n = 24$)			
	30 min		60 min		30 min		60 min	
	Align.	RMSE	Align.	RMSE	Align.	RMSE	Align.	RMSE
DOCAS (20 $\times$ budget)	1.51 $\pm$ 0.82	22.59 $\pm$ 2.62	2.75 $\pm$ 0.77	35.80 $\pm$ 3.64	1.64 $\pm$ 0.84	22.20 $\pm$ 3.33	3.64 $\pm$ 2.22	32.82 $\pm$ 5.21
Budget 0.1 $\times$	9.23 $\pm$ 3.21	21.87 $\pm$ 3.00	20.41 $\pm$ 8.08	34.47 $\pm$ 4.15	8.21 $\pm$ 3.28	21.64 $\pm$ 3.44	16.29 $\pm$ 6.88	31.78 $\pm$ 5.47
Budget 1 $\times$	4.62 $\pm$ 1.13	22.08 $\pm$ 2.87	9.50 $\pm$ 2.54	34.99 $\pm$ 3.86	4.48 $\pm$ 1.70	21.78 $\pm$ 3.38	8.12 $\pm$ 3.10	32.20 $\pm$ 5.44
Budget 100 $\times$	0.82 $\pm$ 0.42	22.69 $\pm$ 2.59	2.13 $\pm$ 0.96	35.96 $\pm$ 3.57	0.84 $\pm$ 0.58	22.32 $\pm$ 3.28	2.32 $\pm$ 1.85	33.11 $\pm$ 5.06
Uniform IOB only	7.37 $\pm$ 4.26	22.48 $\pm$ 2.65	11.13 $\pm$ 6.74	35.73 $\pm$ 3.62	3.31 $\pm$ 1.23	22.12 $\pm$ 3.35	7.27 $\pm$ 3.56	32.61 $\pm$ 5.30
DRC applied after prediction	0.00 $\pm$ 0.00	22.79 $\pm$ 2.59	0.00 $\pm$ 0.00	39.03 $\pm$ 5.36	0.00 $\pm$ 0.00	22.45 $\pm$ 3.21	0.00 $\pm$ 0.00	36.81 $\pm$ 8.38
DRC at 0.5 $\times$ S_I	1.57 $\pm$ 0.91	22.58 $\pm$ 2.72	2.68 $\pm$ 0.79	35.23 $\pm$ 3.85	1.66 $\pm$ 0.94	22.17 $\pm$ 3.35	3.35 $\pm$ 2.35	32.56 $\pm$ 5.30
DRC at 2 $\times$ S_I	1.48 $\pm$ 0.67	22.67 $\pm$ 2.44	4.48 $\pm$ 2.85	36.88 $\pm$ 3.83	1.61 $\pm$ 0.73	22.31 $\pm$ 3.29	4.57 $\pm$ 2.77	33.30 $\pm$ 5.26
Clinical DRC (pump settings)	1.58 $\pm$ 0.83	22.90 $\pm$ 2.26	2.70 $\pm$ 0.75	35.72 $\pm$ 3.44	1.99 $\pm$ 1.03	22.60 $\pm$ 3.21	4.56 $\pm$ 3.54	33.54 $\pm$ 5.60

Compared with the baseline, DOCAS reduced AE on OhioT1DM from 12.70 to 1.44 mg/dL at 30 minutes and from 38.13 to 2.79 mg/dL at 60 minutes (RMSE +3.2% and +4.4%), and on AZT1D from 19.62 to 1.70 mg/dL and from 49.30 to 4.13 mg/dL (RMSE +2.7% and +3.0%). With twelve five-minute lags per channel, RMSE decreased for both arms and the DOCAS AE decreased even further (0.58 vs. 1.44 mg/dL at 30 minutes on OhioT1DM). In the ReplayBG evaluation, the ceteris paribus plot of the baseline turned upward beyond 2.5 U and the corresponding DSS underdosed. The aligned forecaster, in contrast, stayed within 1.83 mg/dL of the target curve, recommended 2.23 instead of 1.09 U on average, and increased the post-decision time in range from 73.61% to 76.01% with no time below range under the safety guard (Table 3, Figure 5).

Alignment error and synthetic data. A forecaster used for dosing has to answer how glucose changes if the dose changes and nothing else does, and the RMSE does not test this. In this study, every unaligned model responded to insulin with a flat or wrong-signed curve, including models with competitive RMSE. This can already be seen in the ceteris paribus plots. The AE, however, quantifies it, so that the insulin response can be reported next to the RMSE, compared across models and cohorts, and tracked when the synthetic budget changes (Tables 2 and 3). Since DOCAS adds the target curve to the training data and not to the model, the same procedure could be used for CatBoost and an MLP without a custom loss, layer, or mechanistic component. A monotonic constraint would only have fixed the sign, and a monotone ceteris paribus plot that is too flat or too steep still leads to under- or overdosing [2, 32, 43]. It is important to note that the target DRC and the associated AE by no means measure a deviation from an absolute ground truth. The ceteris paribus plots of the patient's forecaster are compared with the same UVa/Padova curve, which describes the expected insulin action in a population and not in the individual patient [8, 9]. We chose a population curve because an individual dose–response curve is hard to obtain. Insulin sensitivity varies between people and, within a person, with time of day and physical activity [44–46]. The observational logs cannot provide such a curve, since they are confounded in the way described above, and estimating it directly would require controlled dosing experiments that are rarely feasible in routine care. The population curve is therefore a distant, pharmacology-based reference, but aligning a forecaster to it may still be better than no alignment at all, since an unaligned forecaster can respond to insulin with the wrong sign. Accordingly, a low AE shows that the forecaster responds to insulin as described by the population curve, but not that the pharmacology of the individual patient has been recovered.

Relation to mechanistic, hybrid, and mixed-effects models. Using a population mechanistic model as physiological basis for a data-driven model is an established strategy [47], and DOCAS follows this strategy: the biology comes from UVa/Padova and is not learned from the logs. Synthetic labels, explicit integration, and a physics-informed loss can all encode this prior knowledge, but only explicit integration keeps the equations in the model and thereby the ability to extrapolate [33–35]. Mechanistic, hybrid, and NONMEM-type mixed-effects models answer ceteris paribus dose questions within their structure and quantify the variability between subjects [48–50]. They are the better choice when the equations must hold exactly and have to extrapolate. DOCAS addresses a different setting, in which a data-driven forecaster trained on the patient's logs is used by a decision support system and its insulin response has to be checked and corrected. The DRC describes the population, whereas the observations are the patient's own glucose, meal, and insulin records. DOCAS combines both by training one model on the union of observed and synthetic rows. In this way, the population curve determines the insulin response, while the observations shape everything else the forecaster learns about the patient. Since the prior knowledge enters only through training rows that compete with the observed

data, the aligned *ceteris paribus* plot never matches the target exactly. The wrapper ablation shows the benefit of this approach: applying the DRC after prediction, on top of a forecaster trained without insulin, matches the curve exactly ($AE = 0$) but increases the 60-minute RMSE from 35.80 to 39.03 mg/dL on OhioT1DM and from 32.82 to 36.81 mg/dL on AZT1D (Table 4). Learning from the observations and the population curve together therefore results in a better fit of the patient’s glucose than adding the curve afterwards.

Limitations. In the main study, only one target was used, the UVa/Padova DRC derived from ReplayBG. The same curve was used to label the synthetic rows and to compute the AE, and the in-silico decision support evaluation was run in ReplayBG, whose simulated patients follow the same model. The evaluation is therefore not independent of the target: a forecaster aligned to this curve is scored against the curve it was trained on and tested in a simulator built on the same pharmacology. We addressed this limitation directly with a second, realistic target DRC based on each participant’s pump settings, i.e. the correction factor and the insulin action curve [51]. DOCAS trained on this pump-settings target still aligned and still improved decision support (Section 4.4, Appendix D), so the benefits are not tied to the UVa/Padova DRC. In-silico dosing was only evaluated on open-loop OhioT1DM records, since closed-loop automated insulin delivery in AZT1D continuously adjusts basal rates and delivers automated boluses, which confounds an external DSS evaluation. Furthermore, the absence of time below range depends on the explicit hypoglycaemia safety guard (Appendix C). Single-horizon forecasters evaluate glucose at 60 minutes and thus capture only $\approx 20\%$ of rapid-acting insulin action, which makes an external constraint against late nadirs necessary. Finally, the in-silico replay covered 11 postprandial windows across four eligible participants and therefore provides proof-of-concept evidence rather than the evidence of a clinical trial.

Future directions. Our first aim is a better trade-off between alignment and accuracy (Section 4.2). Better sampling of the synthetic rows could bring the *ceteris paribus* plots closer to the target without any cost in RMSE, or even with an improvement. Compared with the UVa/Padova simulator itself, the aligned forecaster predicts the next hour of glucose from the patient’s own logs while responding to insulin as described by the population curve. A next step would be to replace the population curve with the dose–response of each individual patient, estimated from the logs after adjusting for meals, time of day, and activity [3, 30, 44, 46]. Beyond insulin, the approach can be applied to other treatments whenever a population dose–response for that input can be stated. An open-source implementation is available at <https://github.com/Prgrmmrjns/docas>.

6. Conclusion

A glucose forecaster that is used to choose insulin doses should not only predict glucose accurately but also respond correctly when the dose changes. The held-out error only evaluates the former. In this study, we used the alignment error to evaluate the latter by comparing the insulin response of the forecaster with a population dose–response curve, and we introduced DOCAS, which corrects this response by adding synthetic training rows without changing the forecaster or replacing it with a mechanistic model. On two type 1 diabetes cohorts, the aligned forecasters followed the target curve at a small cost in accuracy. In silico, they recommended larger and more effective correction boluses that increased the time in range without hypoglycaemia under a safety guard.

Data availability

OhioT1DM is distributed by the Ohio University College of Engineering under a data-use agreement [39]. AZT1D is publicly available as described by Khamesian et al. [41, 42]. Processed features and result JSONs are reproduced from the accompanying code.

Code availability

Source code to reproduce all tables and figures is available at <https://github.com/Prgrmmrjns/DOCAS-manuscript>. An implementation of the alignment procedure is available at <https://github.com/Prgrmmrjns/docas>.

A. Supplementary: target DRC and study data

OhioT1DM and AZT1D share one target DRC, obtained by simulating a rapid-acting correction bolus against a matched no-bolus control from basal steady state in ReplayBG [3] and recording the endpoint interstitial lowering at

horizon H . Let $IG^u(H)$ and $IG^{\text{ctl}}(H)$ denote interstitial glucose under dose u and under the control. The fractional lowering and ambient-scaled target DRC are

$$R_H(u) = \frac{1}{G_0} (IG^{\text{ctl}}(H) - IG^u(H)), \quad (2)$$

$$\tau_H(u; g) = -g R_H(u).$$

Reference glucose is $G_0=220$ mg/dL. Body weight is 75 kg. Insulin sensitivity is the mode of ReplayBG’s prior, $S_I=1.15 \times 10^{-3}$ mL/(μ U·min) (Gamma shape $\alpha=3.3$, scale 5×10^{-4}). The remaining simulator values are the published prior modes: glucose effectiveness $S_G=0.017$, insulin-action coupling $p_2=0.11$, subcutaneous rates $k_d=0.021$ min $^{-1}$ and $k_{a2}=0.011$ min $^{-1}$, elimination $k_e=0.127$ min $^{-1}$, volumes $V_I=0.135$ dL/kg and $V_G=1.45$ dL/kg, and basal setpoint $G_b=119$ mg/dL. Bolus lag is 8 min on both cohorts. Sensor smoothing follows the device: Enlite on OhioT1DM ($\alpha=10$) and Dexcom G6 on AZT1D ($\alpha=5$). Figure 2 plots the simulated drop and the resulting target DRC.

For forecasting, $u=1$ is each patient’s training-maximum bolus insulin on board. That span is often longer on AZT1D (closed-loop automated delivery) than on OhioT1DM (open-loop pump therapy). The ReplayBG decision-support search evaluates 0–5 U. Cohort splits and features are in Section 3.4.

B. Supplementary: MLP and lag features

Table 5 repeats Table 2 for a one-hidden-layer MLP (256 ReLU units, Adam, 100 epochs, learning rate 0.03, batch size 512; five seeds) and for CatBoost with twelve five-minute lags per channel (36 inputs; 60 min history; one seed). The MLP insulin ICEs track the stated DRC at a small RMSE cost on both cohorts. The observational MLP starts further from the target than CatBoost, especially on AZT1D at 60 min, where long insulin-on-board spans stretch the normalised probe axis, and DOCAS closes that gap to the same few mg/dL. With the lags, RMSE falls for both arms, the baseline stays far from the stated DRC, and DOCAS alignment error is smaller than with instantaneous inputs.

Table 5

MLP (five seeds) and CatBoost with twelve five-minute lags (one seed). Same protocol as Table 2. Cohort mean $\pm$ SD. Bold*: significantly better arm (Section 3.4).

Approach	30 min		60 min	
	RMSE	Align.	RMSE	Align.
OhioT1DM ($n = 12$)				
MLP baseline	22.23 $\pm$ 3.07	20.93 $\pm$ 8.48	35.41 $\pm$ 4.54	49.29 $\pm$ 25.25
MLP DOCAS	22.74 $\pm$ 2.62	1.89 $\pm$ 0.56*	36.33 $\pm$ 3.32	4.91 $\pm$ 0.75*
Lags baseline	20.15 $\pm$ 2.36*	10.33 $\pm$ 6.83	32.88 $\pm$ 4.34*	42.97 $\pm$ 21.29
Lags DOCAS	21.82 $\pm$ 2.22	0.58 $\pm$ 0.17*	36.03 $\pm$ 3.88	2.36 $\pm$ 0.72*
AZT1D ($n = 24$)				
MLP baseline	21.96 $\pm$ 3.60*	39.74 $\pm$ 33.91	32.51 $\pm$ 5.66*	92.38 $\pm$ 69.86
MLP DOCAS	22.41 $\pm$ 3.24	2.14 $\pm$ 0.69*	33.42 $\pm$ 4.86	4.70 $\pm$ 1.17*
Lags baseline	20.12 $\pm$ 3.50*	15.89 $\pm$ 15.65	30.44 $\pm$ 5.27*	46.42 $\pm$ 26.32
Lags DOCAS	21.89 $\pm$ 3.83	0.74 $\pm$ 0.40*	33.34 $\pm$ 6.19	2.03 $\pm$ 0.72*

C. Supplementary: hypoglycaemia guard

As in Section 4.3, each candidate bolus on 0–5 U is scored on the ICE. Before a dose is suggested, the 60-minute predicted fall is scaled by the insulin still to act by 180 minutes (about fourfold on the same insulin-action curve) and doubled, and the implied nadir must stay above 70 mg/dL; the doubling is a forecast-optimism margin. A rejected candidate falls back to the next smaller admissible dose; both DSS arms are checked identically. Figure 6 shows the loop on patient 588.

Table 6 removes the guard on the same windows, twins, cost, target, trigger, penalty, and search grid. The observational arm also moves (1.09 $\rightarrow$ 1.32 U, post-decision TIR 73.61 $\rightarrow$ 73.86%). DOCAS insulin rises from 2.23 to 2.91 U; post-decision TIR is nearly unchanged (76.01 $\rightarrow$ 75.76%) but TBR appears (1.26%), all from patient 588 at start 1962 (Figure 5: 3.5 $\rightarrow$ 6.5 U, 13.89% below range after the first decision). That twin sits at 1.30 $\times$ the prior

Insulin dose–response in type 1 diabetes

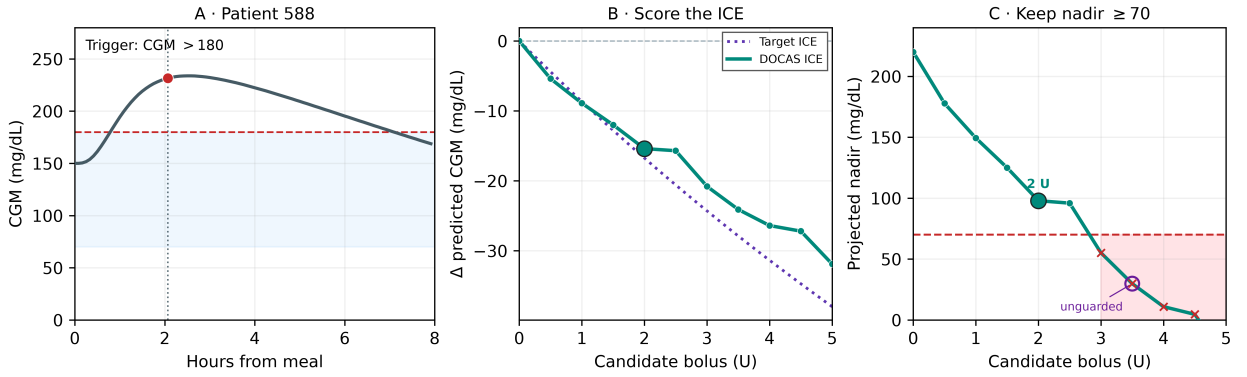


Figure 6: Hypoglycaemia-guard loop on OhioT1DM patient 588. **A:** hyperglycaemia trigger (CGM > 180 mg/dL, red dashed) at the decision time (vertical dotted). **B:** each candidate bolus scored on the ICE against the target DRC. **C:** projected worst-case nadir versus dose; crosses below 70 mg/dL are inadmissible. Filled marker: guarded pick; open marker: unguarded pick on the same ICE.

modal S_I , inside the $2\times$ margin; without the margin the rejected dose looks safe (nadir 125 mg/dL) even though the twin drops below 70 mg/dL. The guard corrects optimism of a 60-minute cost that has not yet seen the nadir. Alignment does not by itself limit how much insulin is given; a trajectory forecaster could penalise late hypoglycaemia directly.

Table 6

The same 11 OhioT1DM windows from 4 participants and the same CIB cost as Table 3, but without the hypoglycaemia guard (no 70 mg/dL floor, no 2× margin), i.e. Prendin-faithful scoring. TIR, TBR, and TAR are post-decision. Mean $\pm$ SD.

Metric	No DSS	Baseline	DOCAS
TBR (%)	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	1.26 $\pm$ 4.19
TIR (%)	61.49 $\pm$ 34.22	73.86 $\pm$ 23.92	75.76 $\pm$ 23.12
TAR (%)	38.51 $\pm$ 34.22	26.14 $\pm$ 23.92	22.98 $\pm$ 21.11
Insulin (U)	0.00 $\pm$ 0.00	1.32 $\pm$ 1.35	2.91 $\pm$ 2.77

D. Supplementary: a pump-settings target DRC

The alignment follows whichever dose–response is supplied, so this appendix repeats it on a second curve taken from each participant’s pump settings rather than from the UVa/Padova DRC used in the main study. The last ablation row of Table 4 replaces the UVa/Padova target DRC with a pump-settings curve [51]:

$$\tau_H(u; g) = -\frac{1800}{\text{TDD}} u A(H) \frac{g}{G_0}, \quad (3)$$

where 1800/TDD is the participant’s correction factor (mg/dL per unit at $G_0=220$ mg/dL; OhioT1DM median 34, range 17–74; AZT1D 51, 15–100) and $A(H)$ is the insulin-action share by horizon H ($t_p=75$ min, $t_d=300$ min; $A(30)=0.075$, $A(60)=0.236$). The product is linear in dose and personal rather than cohort-wide. At G_0 a unit lowers glucose by 2.69 and 8.46 mg/dL on average in OhioT1DM at 30 and 60 min (3.87 and 12.17 in AZT1D). Against the UVa/Padova curve (0.74 and 8.25 mg/dL per unit) the two constructions nearly coincide at 60 min but disagree at 30 min, where UVa/Padova puts more delay into early insulin action.

TDD is mean daily basal plus bolus. On AZT1D, basal rates above 5 U/h are excluded and TDD is clipped to 18–120 U/day (correction factor 100–15 mg/dL per unit). Forecasting results are in Table 4.

Table 7 repeats the ReplayBG protocol on the same eleven windows with DOCAS retrained on Eq. 3. Twins, screens, baseline, and UVa/Padova DOCAS are unchanged. Post-decision TIR rises to 75.88% (vs. 73.61% Baseline, 76.01% UVa/Padova DOCAS) on 2.36 U with zero TBR under the same guard. The dosing gain is still there when the supplied curve is this pump-settings DRC.

Table 7

Same 11 OhioT1DM windows and hypoglycaemia guard as Table 3 (4 participants), with DOCAS retrained on each patient’s pump-settings DRC. Study-DRC DOCAS columns are the main-text arm. TIR, TBR, and TAR are post-decision. Mean $\pm$ SD.

Metric	No DSS	Baseline	DOCAS (study DRC)	DOCAS (pump-settings DRC)
TBR (%)	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
TIR (%)	61.49 $\pm$ 34.22	73.61 $\pm$ 23.83	76.01 $\pm$ 21.95	75.88 $\pm$ 21.91
TAR (%)	38.51 $\pm$ 34.22	26.39 $\pm$ 23.83	23.99 $\pm$ 21.95	24.12 $\pm$ 21.91
Insulin (U)	0.00 $\pm$ 0.00	1.09 $\pm$ 1.41	2.23 $\pm$ 2.04	2.36 $\pm$ 2.58

References

- [1] F. Prendin, J. Pavan, G. Cappon, S. Del Favero, G. Sparacino, A. Facchinetti, The importance of interpreting machine learning models for blood glucose prediction in diabetes: an analysis using SHAP, *Scientific Reports* 13 (2023) 16865.
- [2] T. Kushner, S. Sankaranarayanan, M. D. Breton, Conformance verification for neural network models of glucose–insulin dynamics, in: *Proceedings of the 23rd ACM International Conference on Hybrid Systems: Computation and Control*, 2020, pp. 1–12. doi:10.1145/3365365.3382210.
- [3] G. Cappon, M. Vettoretti, G. Sparacino, S. Del Favero, A. Facchinetti, ReplayBG: a digital twin-based methodology to identify a personalized model from type 1 diabetes data and simulate glucose concentrations to assess alternative therapies, *IEEE Transactions on Biomedical Engineering* 70 (2023) 3205–3216.
- [4] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, F. A. Wichmann, Shortcut learning in deep neural networks, *Nature Machine Intelligence* 2 (2020) 665–673.
- [5] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2 ed., Cambridge University Press, 2009. doi:10.1017/CB09780511803161.

- [6] B. Schölkopf, F. Locatello, S. Bauer, N. R. Ke, N. Kalchbrenner, A. Gretton, J. v. Kügelgen, Toward causal representation learning, *Proceedings of the IEEE* 109 (2021) 612–634.
- [7] J. Pearl, The seven tools of causal inference, with reflections on machine learning, *Communications of the ACM* 62 (2019) 54–60.
- [8] C. Dalla Man, R. A. Rizza, C. Cobelli, Meal simulation model of the glucose-insulin system, *IEEE Transactions on Biomedical Engineering* 54 (2007) 1740–1749.
- [9] C. Cobelli, E. Renard, B. Kovatchev, Diabetes: models, signals, and control, *IEEE Reviews in Biomedical Engineering* 2 (2009) 54–96.
- [10] P. Biecek, T. Burzykowski, *Explanatory Model Analysis: Explore, Explain, and Examine Predictive Models*, Chapman and Hall/CRC, 2021. doi:10.1201/9780429027192.
- [11] A. Goldstein, A. Kapelner, J. Bleich, E. Pitkin, Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation, *Journal of Computational and Graphical Statistics* 24 (2015) 44–65.
- [12] J. H. Friedman, Greedy function approximation: a gradient boosting machine, *The Annals of Statistics* 29 (2001) 1189–1232.
- [13] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [14] M. T. Ribeiro, S. Singh, C. Guestrin, "Why should I trust you?": Explaining the predictions of any classifier, in: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144. doi:10.1145/2939672.2939778.
- [15] C.-K. Yeh, C.-Y. Hsieh, A. Suggala, D. I. Inouye, P. Ravikumar, On the (in)fidelity and sensitivity of explanations, in: *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [16] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, B. Kim, Sanity checks for saliency maps, in: *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [17] W. Samek, A. Binder, G. Montavon, S. Lapuschkin, K.-R. Müller, Evaluating the visualization of what a deep neural network has learned, *IEEE Transactions on Neural Networks and Learning Systems* 28 (2017) 2660–2673.
- [18] M. Nauta, J. Trienes, S. Pathak, E. Nguyen, M. Peters, Y. Schmitt, J. Schlötterer, M. van Keulen, C. Seifert, From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable AI, *ACM Computing Surveys* 55 (2023) 295:1–295:42.
- [19] Y. Jung, S. Kasiviswanathan, J. Tian, D. Janzing, P. Blöbaum, E. Bareinboim, On measuring causal contributions via do-interventions, in: *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, 2022, pp. 10476–10501. URL: <https://proceedings.mlr.press/v162/jung22a.html>.
- [20] T. Heskes, E. Sijben, I. G. Bucur, T. Claassen, Causal Shapley values: Exploiting causal knowledge to explain individual predictions of complex models, in: *Advances in Neural Information Processing Systems*, volume 33, 2020, pp. 4778–4789.
- [21] S. Lapuschkin, S. Wäldchen, A. Binder, G. Montavon, W. Samek, K.-R. Müller, Unmasking Clever Hans predictors and assessing what machines really learn, *Nature Communications* 10 (2019) 1096.
- [22] A. S. Ross, M. C. Hughes, F. Doshi-Velez, Right for the right reasons: Training differentiable models by constraining their explanations, in: *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence*, 2017, pp. 2662–2670. doi:10.24963/ijcai.2017/371.
- [23] G. Erion, J. D. Janizek, P. Sturmfels, S. M. Lundberg, S.-I. Lee, Improving performance of deep learning models with axiomatic attribution priors and expected gradients, *Nature Machine Intelligence* 3 (2021) 620–631.
- [24] L. Rieger, C. Singh, W. J. Murdoch, B. Yu, Interpretations are useful: Penalizing explanations to align neural networks with prior knowledge, in: *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, 2020, pp. 8116–8126. URL: <https://proceedings.mlr.press/v119/rieger20a.html>.
- [25] R. R. Selvaraju, S. Lee, Y. Shen, H. Jin, S. Ghosh, L. Heck, D. Batra, D. Parikh, Taking a HINT: Leveraging explanations to make vision and language models more grounded, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 2591–2600. doi:10.1109/ICCV.2019.00268.
- [26] H. Zhang, M. Cisse, Y. N. Dauphin, D. Lopez-Paz, mixup: Beyond empirical risk minimization, in: *International Conference on Learning Representations*, 2018.
- [27] D. Kaushik, E. Hovy, Z. C. Lipton, Learning the difference that makes a difference with counterfactually-augmented data, in: *International Conference on Learning Representations*, 2020.
- [28] J. Noguer, I. Contreras, O. Mujahid, A. Beneyto, J. Vehi, Generation of individualized synthetic data for augmentation of the type 1 diabetes data sets using deep learning models, *Sensors* 22 (2022) 4944.
- [29] T. Zhu, K. Li, P. Herrero, P. Georgiou, GluGAN: Generating personalized glucose time series using generative adversarial networks, *IEEE Journal of Biomedical and Health Informatics* 27 (2023) 5122–5133.
- [30] F. Prendin, A. Facchinetti, G. Cappon, Data augmentation via digital twins to develop personalized deep learning glucose prediction algorithms for type 1 diabetes in poor data context, *IEEE Transactions on Biomedical Engineering* 73 (2026) 2496–2506.
- [31] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, LightGBM: a highly efficient gradient boosting decision tree, in: *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [32] D. Runje, S. M. Shankaranarayana, Constrained monotonic neural networks, in: *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, 2023, pp. 29338–29353. URL: <https://proceedings.mlr.press/v202/runje23a.html>.
- [33] M. von Stosch, R. Oliveira, J. Peres, S. Feyo de Azevedo, Hybrid semi-parametric modeling in process systems engineering: Past, present and future, *Computers & Chemical Engineering* 60 (2014) 86–101.
- [34] B. Fiedler, A. Schuppert, Local identification of scalar hybrid models with tree structure, *IMA Journal of Applied Mathematics* 73 (2008) 449–476.
- [35] M. E. Samadi, S. Kiefer, S. J. Fritsch, J. Bickenbach, A. Schuppert, A training strategy for hybrid models to break the curse of dimensionality, *PLOS ONE* 17 (2022) e0274569.

- [36] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *Journal of Computational Physics* 378 (2019) 686–707.
- [37] A. Esponera, G. Cinà, Inducing causal structure applied to glucose prediction for T1DM patients, in: *Proceedings of the Fourth Conference on Causal Learning and Reasoning*, volume 275 of *Proceedings of Machine Learning Research*, 2025, pp. 920–946. URL: <https://proceedings.mlr.press/v275/esponera25a.html>.
- [38] R. N. Bergman, Y. Z. Ider, C. R. Bowden, C. Cobelli, Quantitative estimation of insulin sensitivity, *American Journal of Physiology-Endocrinology and Metabolism* 236 (1979) E667–E677.
- [39] C. Marling, R. Bunesco, The OhioT1DM dataset for blood glucose level prediction: update 2020, in: *The 5th International Workshop on Knowledge Discovery in Healthcare Data*, volume 2675 of *CEUR Workshop Proceedings*, 2020, pp. 71–74.
- [40] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, A. Gulin, CatBoost: unbiased boosting with categorical features, in: *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [41] S. Khamesian, A. Arefeen, B. M. Thompson, A. Grando, H. Ghasemzadeh, AZT1D: A real-world dataset for type 1 diabetes, in: *2025 IEEE-EMBS International Conference on Body Sensor Networks (BSN)*, 2025. doi:10.1109/BSN66969.2025.11337496.
- [42] S. Khamesian, A. Arefeen, B. M. Thompson, A. Grando, H. Ghasemzadeh, AZT1D: A real-world dataset for type 1 diabetes, Mendeley Data, 2025. doi:10.17632/gk9m674wcx.1.
- [43] J. C. Wolber, M. E. Samadi, J. Sellin, A. Schuppert, Multimodal large language models and mechanistic modeling for glucose forecasting in type 1 diabetes patients, *Journal of Biomedical Informatics* 172 (2025) 104945.
- [44] L. Hinshaw, C. Dalla Man, D. K. Nandy, A. Saad, A. E. Bharucha, J. A. Levine, R. A. Rizza, R. Basu, C. Cobelli, A. Basu, Diurnal pattern of insulin action in type 1 diabetes: Implications for a closed-loop system, *Diabetes* 62 (2013) 2223–2229.
- [45] R. Visentin, C. Dalla Man, Y. C. Kudva, A. Basu, C. Cobelli, Circadian variability of insulin sensitivity: Physiological input for in silico artificial pancreas, *Diabetes Technology & Therapeutics* 17 (2015) 1–7.
- [46] M. C. Riddell, I. W. Gallen, C. E. Smart, C. E. Taplin, P. Adolfsson, A. N. Lumb, A. Kowalski, R. Rabasa-Lhoret, R. J. McCrimmon, C. Hume, F. Annan, P. A. Fournier, C. Graham, B. Bode, P. Galassetti, T. W. Jones, I. San Millán, T. Heise, A. L. Peters, A. Petz, L. M. Laffel, Exercise management in type 1 diabetes: A consensus statement, *The Lancet Diabetes & Endocrinology* 5 (2017) 377–390.
- [47] H. Fröhlich, R. Balling, N. Beerenwinkel, O. Kohlbacher, S. Kumar, T. Lengauer, M. H. Maathuis, Y. Moreau, S. A. Murphy, T. M. Przytycka, M. Rebhan, H. Röst, A. Schuppert, M. Schwab, R. Spang, D. Stekhoven, J. Sun, A. Weber, D. Ziemek, B. Zupan, From hype to reality: Data science enabling personalized medicine, *BMC Medicine* 16 (2018) 150.
- [48] L. B. Sheiner, S. L. Beal, Evaluation of methods for estimating population pharmacokinetic parameters. I. Michaelis–Menten model: Routine clinical pharmacokinetic data, *Journal of Pharmacokinetics and Biopharmaceutics* 8 (1980) 553–571.
- [49] D. R. Mould, R. N. Upton, Basic concepts in population modeling, simulation, and model-based drug development, *CPT: Pharmacometrics & Systems Pharmacology* 1 (2012) e6.
- [50] H. M. Jones, K. Rowland-Yeo, Basic concepts in physiologically based pharmacokinetic modeling in drug discovery and development, *CPT: Pharmacometrics & Systems Pharmacology* 2 (2013) e63.
- [51] J. Walsh, R. Roberts, T. Bailey, Guidelines for optimal bolus calculator settings in adults, *Journal of Diabetes Science and Technology* 5 (2011) 129–135.